WORKED SOLUTION

STEP III 2000, Question 8

Pure • Recurrence Relations

STEP III 2000, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q8

The key idea

The defining rule is non-linear and involves division, so it is awkward to work with directly. The trick is to rewrite it as a product, $a_n a_{n-2} - a_{n-1}^2 = 1$. This says a certain combination of three consecutive terms is always 1. Writing the same fact one step further on and subtracting the two versions turns the non-linear rule into the linear one. Once the recurrence is linear, the closed form is the standard second-order recurrence machinery: roots of the auxiliary quadratic, then matching two starting values.

Solution

A preliminary: the sequence is well defined

Each step divides by a_{n-2}, so we should make sure no term is zero. In fact every term is positive. We have $a_0 = a_1 = 1 > 0$. If a_{n-1} and a_{n-2} are positive, then $1 + a_{n-1}^2 > 0$ and $a_{n-2} > 0$, so $a_n > 0$. By induction, $a_n > 0$ for all n. The first few terms are

$$a_0 = 1, \quad a_1 = 1, \quad a_2 = \frac{1+1}{1} = 2, \quad a_3 = \frac{1+4}{1} = 5, \quad a_4 = \frac{1+25}{2} = 13.$$

The linear recurrence

Multiplying the definition by a_{n-2} gives, for every $n \geq 2$,

$$a_n a_{n-2} = 1 + a_{n-1}^2. \qquad (*)$$

Let $P(k)$ be the statement $a_k = 3a_{k-1} - a_{k-2}$. We prove $P(k)$ for all $k \geq 2$ by induction on k.

Base case. $a_2 = 2$ and $3a_1 - a_0 = 3 - 1 = 2$, so $P(2)$ is true.

Inductive step. Suppose $P(k)$ is true for some $k \geq 2$, so that

$$a_k + a_{k-2} = 3a_{k-1}.$$

We need to show $a_{k+1} = 3a_k - a_{k-1}$. By definition, $a_{k+1} = (1 + a_k^2)/a_{k-1}$. The idea is to replace the awkward 1 in the numerator using $(*)$ with $n = k$, which says $1 = a_k a_{k-2} - a_{k-1}^2$:

$$\begin{aligned} a_{k+1} &= \frac{a_k a_{k-2} - a_{k-1}^2 + a_k^2}{a_{k-1}} \\ &= \frac{a_k\,(a_k + a_{k-2}) - a_{k-1}^2}{a_{k-1}} \\ &= \frac{a_k \cdot 3a_{k-1} - a_{k-1}^2}{a_{k-1}} && \text{by } P(k) \\ &= 3a_k - a_{k-1} && \text{since } a_{k-1} > 0. \end{aligned}$$

So $P(k+1)$ is true. By induction, $a_n = 3a_{n-1} - a_{n-2}$ for all $n \geq 2$.

Ask yourself Where did the inductive hypothesis come in, and why is $(*)$ at $n = k$ fair to use?

The hypothesis was used once, to replace $a_k + a_{k-2}$ by $3a_{k-1}$. That bracket only appears because we swapped the 1 for $a_k a_{k-2} - a_{k-1}^2$. Using $(*)$ needs no hypothesis at all: it is just the definition of a_k multiplied out, valid for every $k \geq 2$. In other words, the quantity $a_n a_{n-2} - a_{n-1}^2$ is the same at every step, and that is what carries the linear relation forward.

The closed form

Write $\beta = \alpha^{-1}$. Since $\alpha = \frac{1+\sqrt{5}}{2}$,

$$\beta = \frac{2}{1+\sqrt{5}} = \frac{2(\sqrt{5}-1)}{5-1} = \frac{\sqrt{5}-1}{2}, \qquad \text{so} \qquad \alpha + \beta = \sqrt{5}, \quad \alpha\beta = 1.$$

Hence

$$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 5 - 2 = 3, \qquad \alpha^2\beta^2 = 1,$$

so α^2 and β^2 are the two roots of the auxiliary equation $x^2 - 3x + 1 = 0$ of our recurrence. (Explicitly, they are $\frac{3\pm\sqrt{5}}{2}$.)

Now define, for $n \geq 1$,

$$b_n = \frac{\alpha^{2n-1} + \beta^{2n-1}}{\sqrt{5}}.$$

We show $b_n = a_n$ by checking that b_n has the same first two terms as a_n and obeys the same recurrence.

It obeys the recurrence. If x is a root of $x^2 = 3x - 1$, then multiplying by x^{n-2} gives $x^n = 3x^{n-1} - x^{n-2}$. Since $\alpha^{2n-1} = \beta\,(\alpha^2)^n$ (using $\alpha\beta = 1$), taking $x = \alpha^2$ and multiplying by β gives

$$\alpha^{2n-1} = 3\alpha^{2n-3} - \alpha^{2n-5}.$$

In the same way, with $x = \beta^2$ and multiplying by α, we get $\beta^{2n-1} = 3\beta^{2n-3} - \beta^{2n-5}$. Adding and dividing by $\sqrt{5}$:

$$b_n = 3b_{n-1} - b_{n-2} \qquad (n \geq 3).$$

The starting values agree.

$$b_1 = \frac{\alpha + \beta}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1 = a_1,$$
$$b_2 = \frac{\alpha^3 + \beta^3}{\sqrt{5}} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{\sqrt{5}} = \frac{5\sqrt{5} - 3\sqrt{5}}{\sqrt{5}} = 2 = a_2.$$

Conclusion. Both (a_n) and (b_n) satisfy $u_n = 3u_{n-1} - u_{n-2}$ for $n \geq 3$, and they agree at $n = 1$ and $n = 2$. A recurrence like this determines each term from the two before it, so by (strong) induction $a_n = b_n$ for every $n \geq 1$. That is,

$$a_n = \frac{\alpha^{2n-1} + \alpha^{-(2n-1)}}{\sqrt{5}} \qquad (n \geq 1),$$

as required.

? **Ask yourself** Why start the comparison at $n = 1$ rather than $n = 0$?

The linear recurrence for a_n at $n = 3$ involves a_2 and a_1, so matching b_1 and b_2 is all we need for $n \geq 1$. In fact the formula works at $n = 0$ too: $b_0 = (\alpha^{-1} + \alpha)/\sqrt{5} = 1 = a_0$. So nothing is lost either way.

Checks

- The terms are $1, 1, 2, 5, 13, 34, 89, 233, \ldots$, the Fibonacci numbers $F_1, F_3, F_5, \ldots$ (with $a_0 = 1 = F_{-1}$). Each is three times the previous one minus the one before: $3 \cdot 13 - 5 = 34$.
- Check $(*)$ directly: $34 \cdot 5 - 13^2 = 170 - 169 = 1$.
- Numerically, $a_{10} = 4181$, and $(\alpha^{19} + \alpha^{-19})/\sqrt{5} \approx 4181.000$. The term $\alpha^{-19}/\sqrt{5}$ is tiny, so a_n is the nearest integer to $\alpha^{2n-1}/\sqrt{5}$.
- Computing the sequence exactly from the original rule up to $n = 39$ gives integers satisfying the linear recurrence every time, matching the closed form.

Common slips

- Trying to prove $a_{k+1} = 3a_k - a_{k-1}$ by substituting $a_k = 3a_{k-1} - a_{k-2}$ into $(1 + a_k^2)/a_{k-1}$ and expanding. This just makes a mess. The 1 has to be replaced using the definition one step back.

- Forgetting that the definition divides by a_{n-2}. A complete answer notes that the terms are positive, so the division and the final cancellation of a_{k-1} are legitimate.
- Checking only that b_n satisfies the recurrence, without matching *two* starting values. A second-order recurrence needs both.

Beyond STEP

The constant $a_n a_{n-2} - a_{n-1}^2$ is an instance of Cassini's identity for Fibonacci numbers, $F_{m+1}F_{m-1} - F_m^2 = (-1)^m$. Here we only see odd indices, so the sign is always $+1$. More generally, $u_{n+1}u_{n-1} - u_n^2$ changes by a constant factor at each step for any linear second-order recurrence (it is a 2×2 determinant, and determinants multiply). Rational recurrences that, surprisingly, always produce integers, like this one, are linked to the study of "Somos sequences" and the Laurent phenomenon. For the number theory of Fibonacci-type sequences and continued fractions (note that α is the golden ratio), Davenport's *The Higher Arithmetic* is a lovely read.

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